Gaston Elian Bujia

Ph.D. Candidate in Computer Science (UBA-CONICET) | University Teaching | Data Science and Machine Learning | Computational Neuroscience

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Professional Profile

University teacher and Ph.D. candidate in Computer Science at UBA, with more than ten years of teaching experience across undergraduate and graduate courses. I am currently a doctoral fellow and work at the Applied Artificial Intelligence Lab on computational models of eye movements during visual search in natural scenes. My academic profile combines sustained teaching activity with research in Machine Learning, Bayesian modeling, and computational neuroscience.

Teaching and Academic Background

Teaching

Senior Teaching Assistant / Teaching Assistant Department of Computer Science, School of Exact and Natural Sciences, University of Buenos Aires | *2021 - Present*

- **Senior Teaching Assistant** in **Machine Learning**, appointed through competitive examination, from August 2024 to August 2025.
- **Senior Teaching Assistant** in **Computational Statistics**, appointed through competitive examination, from August 2025 to the present.
- **Teaching Assistant** in **Machine Learning**, appointed through competitive examination, from March 2024 to August 2024.
- **Teaching Assistant** in **Computer Organization**, appointed through competitive examination, from August 2021 to December 2023.
- Previous undergraduate teaching experience in **Mathematics (CBC UBA)** between 2013 and 2021.

Graduate Teaching Assistant Master's Program in Data Mining and Knowledge Discovery, School of Exact and Natural Sciences, University of Buenos Aires | *2021 - 2024*

- Teaching Assistant in **Machine Learning** from 2021 to 2024.
- Graduate-level teaching in the master's program.

Peer Reviewer Frontiers in Psychology / NeurIPS ecosystem | *2021 - 2023*

- Reviewer for **Frontiers in Psychology** (Cognitive Science section) in 2023.
- Reviewer for **NeurIPS 2022 Datasets and Benchmarks**.
- Reviewer for submissions to the **NeurIPS 2021 Shared Visual Representations in Humans and Machines Workshop**.

Thesis Co-supervisor and Undergraduate Research Mentor University of Buenos Aires | *2021 - Present*

- Co-supervision of the Computer Science undergraduate theses of **Fermín Travi** and **Gonzalo Ruarte** (defended in 2023).

- Mentor of the undergraduate research fellowship of **Gianluca Capelo** on individual differences in executive functions (from 2024).
- Mentor of the undergraduate research fellowship of **Fermín Travi** on computational models of human visual search (2021).

Research

Graduate Researcher   Applied Artificial Intelligence Lab (UBA-CONICET) | *2018 - Present*

- **Doctoral work:** computational models of eye movements and visual search in natural scenes under the supervision of Dr. Juan Kamienkowski.
- **Funding:** CONICET internal doctoral fellowship (2018 - 2024).
- Research interests include Machine Learning, computational neuroscience, eye tracking, deep learning, and Bayesian modeling.

Open Research Software and Teaching Resources   GitHub / NeuroLIAA | *2022 - Present*

- Contributor to **ViSioNS**, a benchmark for computational models of human visual search in natural scenes.
- Contributor to **sIBS**, a Bayesian visual-search model for natural scenes.
- Author of the public course repository **Data Processing and Visualization with Python - SAN 2022**.

Professional Activity

Research Data Scientist   Everything ALS | *March 2024 - Present*

- Research Data Scientist working on data-driven analysis in the context of amyotrophic lateral sclerosis.

Education

- **Ph.D. in Computer Science** | UBA (2018 - Present)
  - Doctoral thesis titled *Computational aspects of human vision: prediction of eye movements during visual search.* Expected defense between July and August 2026. Advisor: Dr. Juan Kamienkowski. Co-advisor: Dr. Guillermo Solovey.
- **B.Sc. in Mathematics** | UBA (2018)
  - Applied mathematics focus. Thesis: *Recommendation systems: Factorization Machines and the cold-start problem.* Advisor: Dr. Leandro Lombardi.

Distinctions

- **CONICET Internal Doctoral Fellowship** (2018 - 2024). Advisor: Dr. Juan Kamienkowski.

Selected Scientific Production

1. Ruarte, G., **Bujia, G.**, Care, D., Ison, M. J., & Kamienkowski, J. E. (2025). *Integrating Bayesian and neural networks models for eye movement prediction in hybrid search.* In **Scientific Reports**, 15(1), 16482.

2. Dziemian, S., **Bujia, G.**, Prasse, P., Barańczuk-Turska, Z., Jäger, L. A., Kamienkowski, J. E., & Langer, N. (2025). *Saliency models reveal reduced top-down attention in Attention-Deficit/Hyperactivity Disorder: A naturalistic eye-tracking study.* In **JAACAP Open**, 3(2), 192–204.
3. **Bujia, G.**, Ruarte, G., Sclar, M., Solovey, G., & Kamienkowski, J. E. (2025). *Uncertainty during visual search: Insights from a computational model and behavioral experiment in natural stimuli.* In **bioRxiv**.
4. **Bujia, G.**, Sclar, M., Vita, S., Solovey, G., & Kamienkowski, J. E. (2022). *Modeling human visual search in natural scenes: A combined Bayesian searcher and saliency map approach.* In **Frontiers in Systems Neuroscience**, 16, 882315.
5. Travi, F., Ruarte, G., **Bujia, G.**, & Kamienkowski, J. E. (2022). *ViSioNS: Visual search in natural scenes benchmark.* In **Advances in Neural Information Processing Systems**, 35, 11987–12000.

Technical Skills

- **Scientific Python stack:** Python, NumPy, Pandas, Matplotlib, SciPy, scikit-learn, and Jupyter Notebooks.
- **Machine Learning and modeling:** PyTorch, TensorFlow, Bayesian modeling, eye tracking, and computational modeling.
- **Visualization and teaching materials:** Matplotlib, Seaborn, Plotly, and Jupyter Notebooks as a pedagogical tool.
- **Tools and workflows:** Git, GitHub, LaTeX.
- **Teaching areas:** Machine Learning, computational statistics, computer organization and architecture, undergraduate and graduate instruction.
- **Languages:** Spanish (native), English (professional working proficiency).

Buenos Aires, Argentina